

City of Detroit 2022 - 2025 Sales

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Introduction

This report evaluates the metrics of a linear regression model created from citywide datasets of sales data from the years 2022 - 2024 and 2023 - 2025. Overlapping sales IDs were removed to include only unique sales, leaving a dataset containing single family residential sales from 2022 - 2025. After removing outliers in the initial data prep, a total of 19756 sales were used for model development.

Custom functions were written to automate variable creation from initial datasets, calculate assessing coefficients (PRD, COD, PRB), variance factors, and graphs.

Functions Used: modfunctions_notes.R + modelfunctions.R

R Data File: cod_2022-2025.RData Original Data Files: salesSFApr2023toMarch2025Updated.csv + SFResiSalesJUN2022-MAY2024_w-Exterior1.csv

Model I

Table 1: LN_SPPSF model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19756	19732	24	0.5351	0.1621	0.1611	0.5348	0.4318	165.9671	0	31383.04	31580.32

Table 2: LN_SPPSF model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	3.9561	221.1662	0.0179	0.0000	3.9211	3.9912
single_family1story	0.0509	4.8473	0.0105	0.0000	0.0303	0.0715

term	estimate	statistic	std_error	p_value	conf_low	conf_high
half_duplex1story	0.0816	1.7381	0.0470	0.0822	-0.0104	0.1737
smallest	0.1295	8.6521	0.0150	0.0000	0.1002	0.1588
small	0.0530	4.9302	0.0107	0.0000	0.0319	0.0740
two_plus_baths	0.1798	7.9969	0.0225	0.0000	0.1357	0.2238
powder_room	0.0773	6.1889	0.0125	0.0000	0.0528	0.1018
good	0.3570	12.1469	0.0294	0.0000	0.2994	0.4146
fair	-0.2064	-21.7091	0.0095	0.0000	-0.2250	-0.1877
poor	-0.3558	-16.3789	0.0217	0.0000	-0.3984	-0.3132
very_poor_unsound	-0.3324	-7.3201	0.0454	0.0000	-0.4214	-0.2434
qabove_avg	0.4421	15.5956	0.0284	0.0000	0.3866	0.4977
qbelow_avg	-0.0337	-3.4557	0.0098	0.0005	-0.0528	-0.0146
yb_pre_1929	-0.1747	-15.3768	0.0114	0.0000	-0.1970	-0.1525
yb_1946_to_1960	0.0758	8.1864	0.0093	0.0000	0.0577	0.0940
ln_garagespaces	0.1034	10.4659	0.0099	0.0000	0.0840	0.1227
fireplace	0.0984	10.5351	0.0093	0.0000	0.0801	0.1167
hot_water	0.0727	4.8905	0.0149	0.0000	0.0436	0.1019
elec_baseboard	-0.3213	-2.2362	0.1437	0.0253	-0.6029	-0.0397
wall	-0.1214	-2.5250	0.0481	0.0116	-0.2156	-0.0272
central_air	0.0927	8.6230	0.0107	0.0000	0.0716	0.1137
months_1to9	-0.0102	-4.8981	0.0021	0.0000	-0.0143	-0.0061
months_10to16	0.0187	8.5256	0.0022	0.0000	0.0144	0.0230
months_17to24	0.0099	10.4231	0.0010	0.0000	0.0081	0.0118

Model II

Table 3: LN_PRICE model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19756	19729	27	0.5328	0.3712	0.3703	0.5325	0.4295	447.9058	0	31218.04	31438.99

Table 4: LN_PRICE model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.9441	566.5569	0.0193	0.0000	10.9062	10.9820
ln_landratio	0.2301	13.0188	0.0177	0.0000	0.1955	0.2648
ln_base_bldgsize_ratio	0.9017	22.3717	0.0403	0.0000	0.8227	0.9807
single_family1story	0.0409	3.7183	0.0110	0.0002	0.0193	0.0624
half_duplex1story	-0.1416	-3.0049	0.0471	0.0027	-0.2340	-0.0492
half_duplex2story	-0.1799	-3.9578	0.0454	0.0001	-0.2690	-0.0908
smallest	0.0863	3.8109	0.0226	0.0001	0.0419	0.1307
small	0.0313	2.2824	0.0137	0.0225	0.0044	0.0582
two_plus_baths	0.1701	5.6599	0.0300	0.0000	0.1112	0.2290
powder_room	0.0850	5.3503	0.0159	0.0000	0.0539	0.1162
good	0.3603	12.3082	0.0293	0.0000	0.3029	0.4176
fair	-0.2029	-21.4219	0.0095	0.0000	-0.2215	-0.1844
poor	-0.3507	-16.2074	0.0216	0.0000	-0.3931	-0.3083
very_poor_unsound	-0.3272	-7.2352	0.0452	0.0000	-0.4158	-0.2385
qabove_avg	0.4123	14.4135	0.0286	0.0000	0.3562	0.4683
qbelow_avg	-0.0387	-3.8977	0.0099	0.0001	-0.0581	-0.0192
yb_pre_1929	-0.1449	-12.4982	0.0116	0.0000	-0.1676	-0.1221

term	estimate	statistic	std_error	p_value	conf_low	conf_high
yb_1946_to_1960	0.0593	6.3699	0.0093	0.0000	0.0411	0.0776
ln_garagespaces	0.0815	8.1511	0.0100	0.0000	0.0619	0.1011
fireplace	0.0959	10.1820	0.0094	0.0000	0.0775	0.1144
hot_water	0.0715	4.8188	0.0148	0.0000	0.0424	0.1006
elec_baseboard	-0.3596	-2.5127	0.1431	0.0120	-0.6401	-0.0791
wall	-0.1338	-2.7932	0.0479	0.0052	-0.2277	-0.0399
central_air	0.0910	8.4893	0.0107	0.0000	0.0700	0.1120
months_1to9	-0.0101	-4.8473	0.0021	0.0000	-0.0141	-0.0060
months_10to16	0.0189	8.6121	0.0022	0.0000	0.0146	0.0232
months_17to24	0.0101	10.6913	0.0009	0.0000	0.0083	0.0120

Model III

Table 5: LN_PRICE model (base) assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
19756	0.9849147	1.115474	0.9058778	1.231374	41.59922	-0.5587016

Table 6: LN_PRICE (base) model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19756	19569	187	0.4711	0.5124	0.5077	0.4689	0.3811	110.5488	0	26514.15	27997.7

Table 7: LN_PRICE (+ECFs) model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.3167	519.7985	0.0218	0.0000	11.2740	11.3594
ln_landratio	0.1120	6.2150	0.0180	0.0000	0.0767	0.1474
ln_base_bldgsize_ratio	0.6862	18.4451	0.0372	0.0000	0.6133	0.7591
single_family1story	0.0313	3.1756	0.0099	0.0015	0.0120	0.0507
half_duplex1story	-0.1649	-3.8605	0.0427	0.0001	-0.2486	-0.0812
half_duplex2story	-0.2345	-5.7106	0.0411	0.0000	-0.3150	-0.1540
smallest	0.0444	2.1644	0.0205	0.0304	0.0042	0.0846
small	0.0242	1.9470	0.0124	0.0515	-0.0002	0.0485
two_plus_baths	0.0914	3.2925	0.0278	0.0010	0.0370	0.1458
powder_room	0.0282	1.9551	0.0144	0.0506	-0.0001	0.0566
good	0.2952	11.2329	0.0263	0.0000	0.2437	0.3467
fair	-0.1628	-19.0295	0.0086	0.0000	-0.1796	-0.1460
poor	-0.2918	-15.0666	0.0194	0.0000	-0.3297	-0.2538
very_poor_unsound	-0.2331	-5.7616	0.0405	0.0000	-0.3124	-0.1538
qabove_avg	0.1898	6.3880	0.0297	0.0000	0.1316	0.2481
qbelow_avg	-0.0588	-6.0011	0.0098	0.0000	-0.0780	-0.0396
yb_pre_1929	-0.0741	-6.1807	0.0120	0.0000	-0.0976	-0.0506
yb_1946_to_1960	0.0450	5.0395	0.0089	0.0000	0.0275	0.0625
yb_post_1961	0.0732	2.9722	0.0246	0.0030	0.0249	0.1215
crawl	-0.0798	-3.3483	0.0238	0.0008	-0.1265	-0.0331
slab	-0.1075	-3.8377	0.0280	0.0001	-0.1624	-0.0526
ln_garagespaces	0.1020	11.2718	0.0091	0.0000	0.0843	0.1198
fireplace	0.0626	6.8385	0.0092	0.0000	0.0447	0.0806

term	estimate	statistic	std_error	p_value	conf_low	conf_high
hot_water	0.0423	3.1448	0.0134	0.0017	0.0159	0.0686
wall	-0.1707	-3.7726	0.0453	0.0002	-0.2595	-0.0820
central_air	0.0483	5.0327	0.0096	0.0000	0.0295	0.0672
ecf_1R102	-0.3751	-5.6840	0.0660	0.0000	-0.5044	-0.2457
ecf_1R104	-0.1180	-3.8799	0.0304	0.0001	-0.1776	-0.0584
ecf_1R105	-0.3179	-9.2124	0.0345	0.0000	-0.3855	-0.2502
ecf_1R106	-0.2759	-5.1164	0.0539	0.0000	-0.3817	-0.1702
ecf_1R107	-0.2204	-2.2664	0.0972	0.0234	-0.4110	-0.0298
ecf_1R108	-0.4366	-12.4087	0.0352	0.0000	-0.5056	-0.3676
ecf_1R109	-0.1453	-4.9353	0.0294	0.0000	-0.2030	-0.0876
ecf_1R110	-0.1329	-2.8018	0.0474	0.0051	-0.2258	-0.0399
ecf_1R112	-0.1447	-2.5975	0.0557	0.0094	-0.2539	-0.0355
ecf_1R113	-0.5516	-9.2824	0.0594	0.0000	-0.6681	-0.4351
ecf_1R114	-0.3163	-5.0703	0.0624	0.0000	-0.4386	-0.1941
ecf_1R116	0.1654	4.0654	0.0407	0.0000	0.0857	0.2452
ecf_1R117	-0.1380	-4.0161	0.0344	0.0001	-0.2054	-0.0707
ecf_1R118	-0.2481	-6.0990	0.0407	0.0000	-0.3279	-0.1684
ecf_1R119	-0.3842	-8.4506	0.0455	0.0000	-0.4733	-0.2951
ecf_1R120	-0.1099	-1.8309	0.0600	0.0671	-0.2275	0.0078
ecf_1R121	-0.1947	-5.2589	0.0370	0.0000	-0.2673	-0.1222
ecf_1R122	-0.3847	-9.5320	0.0404	0.0000	-0.4638	-0.3056
ecf_1R123	-0.3310	-6.2254	0.0532	0.0000	-0.4352	-0.2267
ecf_1R124	-0.7050	-9.4355	0.0747	0.0000	-0.8514	-0.5585
ecf_1R126	0.2500	5.6198	0.0445	0.0000	0.1628	0.3372
ecf_1R127	-0.3658	-6.1888	0.0591	0.0000	-0.4816	-0.2499
ecf_1R128	-0.4858	-12.7109	0.0382	0.0000	-0.5607	-0.4109
ecf_1R129	-0.1754	-3.4724	0.0505	0.0005	-0.2745	-0.0764
ecf_1R130	-0.5997	-3.3161	0.1809	0.0009	-0.9542	-0.2452
ecf_1R131	-0.3392	-3.4791	0.0975	0.0005	-0.5303	-0.1481
ecf_1R133	-0.3911	-4.8998	0.0798	0.0000	-0.5475	-0.2346
ecf_1R135	-0.2394	-5.4750	0.0437	0.0000	-0.3252	-0.1537
ecf_1R136	-0.2877	-7.2732	0.0396	0.0000	-0.3652	-0.2102
ecf_1R137	-0.7252	-12.2223	0.0593	0.0000	-0.8415	-0.6089
ecf_2R201	-0.2940	-10.3037	0.0285	0.0000	-0.3499	-0.2380
ecf_2R202	-0.1099	-2.1127	0.0520	0.0346	-0.2119	-0.0079
ecf_2R205	-0.2016	-6.1677	0.0327	0.0000	-0.2656	-0.1375
ecf_2R208	0.4455	6.4056	0.0695	0.0000	0.3092	0.5818
ecf_2R209	0.4115	3.8298	0.1074	0.0001	0.2009	0.6220
ecf_2R210	0.4725	3.3230	0.1422	0.0009	0.1938	0.7511
ecf_2R211	-1.1341	-5.3604	0.2116	0.0000	-1.5487	-0.7194
ecf_2R212	-0.1823	-6.1135	0.0298	0.0000	-0.2408	-0.1239
ecf_2R213	-0.0888	-2.6747	0.0332	0.0075	-0.1540	-0.0237
ecf_2R214	0.0787	3.6075	0.0218	0.0003	0.0359	0.1215
ecf_2R215	0.2982	5.0191	0.0594	0.0000	0.1818	0.4147
ecf_2R218	-0.5561	-7.2378	0.0768	0.0000	-0.7067	-0.4055
ecf_2R219	0.2939	2.2321	0.1317	0.0256	0.0358	0.5519
ecf_2R220	-0.2543	-6.8613	0.0371	0.0000	-0.3269	-0.1816
ecf_2R221	-0.1789	-5.4248	0.0330	0.0000	-0.2435	-0.1143
ecf_2R222	-0.3658	-10.9284	0.0335	0.0000	-0.4314	-0.3002
ecf_2R223	-0.2669	-7.2310	0.0369	0.0000	-0.3393	-0.1946
ecf_2R224	-0.4723	-12.0687	0.0391	0.0000	-0.5491	-0.3956
ecf_2R225	-0.2852	-7.5470	0.0378	0.0000	-0.3593	-0.2112
ecf_2R226	-0.6501	-5.9545	0.1092	0.0000	-0.8641	-0.4361
ecf_2R228	-0.8333	-6.5596	0.1270	0.0000	-1.0823	-0.5843
ecf_2R229	-0.6446	-6.3398	0.1017	0.0000	-0.8439	-0.4453
ecf_2R230	-0.7903	-4.0904	0.1932	0.0000	-1.1691	-0.4116
ecf_3R301	-0.6902	-8.2685	0.0835	0.0000	-0.8538	-0.5266

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_3R302	-0.2823	-1.9733	0.1431	0.0485	-0.5628	-0.0019
ecf_3R303	-0.5966	-18.7380	0.0318	0.0000	-0.6590	-0.5342
ecf_3R304	-0.8435	-13.9468	0.0605	0.0000	-0.9621	-0.7250
ecf_3R305	-0.8727	-7.3268	0.1191	0.0000	-1.1061	-0.6392
ecf_3R307	-0.4622	-14.1768	0.0326	0.0000	-0.5261	-0.3983
ecf_3R308	-0.7535	-10.5121	0.0717	0.0000	-0.8940	-0.6130
ecf_3R309	-0.7655	-10.8343	0.0707	0.0000	-0.9040	-0.6270
ecf_3R310	0.2269	5.2790	0.0430	0.0000	0.1427	0.3112
ecf_3R311	-0.5132	-7.1435	0.0718	0.0000	-0.6540	-0.3724
ecf_3R312	-0.7227	-17.5835	0.0411	0.0000	-0.8033	-0.6422
ecf_3R313	-0.6663	-14.3024	0.0466	0.0000	-0.7576	-0.5750
ecf_3R314	-0.3977	-7.0886	0.0561	0.0000	-0.5077	-0.2877
ecf_3R315	-0.3735	-8.1666	0.0457	0.0000	-0.4631	-0.2838
ecf_3R316	-0.6843	-11.8029	0.0580	0.0000	-0.7980	-0.5707
ecf_3R317	-1.0100	-7.6796	0.1315	0.0000	-1.2678	-0.7522
ecf_3R318	-0.3075	-2.3371	0.1316	0.0194	-0.5654	-0.0496
ecf_3R319	-0.5957	-21.2537	0.0280	0.0000	-0.6506	-0.5408
ecf_3R320	-0.5139	-14.5424	0.0353	0.0000	-0.5832	-0.4447
ecf_3R321	-0.8679	-14.7764	0.0587	0.0000	-0.9831	-0.7528
ecf_3R322	-0.2796	-5.9828	0.0467	0.0000	-0.3712	-0.1880
ecf_3R323	-0.4574	-13.7331	0.0333	0.0000	-0.5226	-0.3921
ecf_3R324	-0.7409	-10.8814	0.0681	0.0000	-0.8743	-0.6074
ecf_3R325	-0.5472	-11.4424	0.0478	0.0000	-0.6409	-0.4535
ecf_3R326	-1.1466	-14.5050	0.0791	0.0000	-1.3016	-0.9917
ecf_4R401	-0.6256	-16.1403	0.0388	0.0000	-0.7016	-0.5497
ecf_4R402	-0.9857	-16.3746	0.0602	0.0000	-1.1036	-0.8677
ecf_4R403	-0.5003	-11.4472	0.0437	0.0000	-0.5860	-0.4146
ecf_4R404	-0.4830	-10.9900	0.0439	0.0000	-0.5691	-0.3968
ecf_4R405	-0.3891	-12.8737	0.0302	0.0000	-0.4484	-0.3299
ecf_4R406	-0.4406	-15.2762	0.0288	0.0000	-0.4971	-0.3841
ecf_4R407	-0.6092	-11.0721	0.0550	0.0000	-0.7171	-0.5014
ecf_4R408	-0.5711	-16.8034	0.0340	0.0000	-0.6377	-0.5045
ecf_4R409	-0.4492	-8.7657	0.0512	0.0000	-0.5497	-0.3488
ecf_4R410	-0.1772	-6.6600	0.0266	0.0000	-0.2294	-0.1251
ecf_4R411	-0.6598	-19.8227	0.0333	0.0000	-0.7250	-0.5945
ecf_4R412	-0.7612	-13.8959	0.0548	0.0000	-0.8686	-0.6538
ecf_4R413	-0.2910	-8.5381	0.0341	0.0000	-0.3578	-0.2242
ecf_4R415	0.1744	3.2948	0.0529	0.0010	0.0707	0.2782
ecf_4R416	-0.7553	-7.9273	0.0953	0.0000	-0.9421	-0.5686
ecf_4R417	-0.1517	-4.5376	0.0334	0.0000	-0.2172	-0.0862
ecf_4R418	-0.4641	-4.1327	0.1123	0.0000	-0.6842	-0.2440
ecf_4R419	-0.6959	-12.1967	0.0571	0.0000	-0.8077	-0.5841
ecf_4R421	-0.7224	-10.0138	0.0721	0.0000	-0.8638	-0.5810
ecf_4R422	0.0975	2.1433	0.0455	0.0321	0.0083	0.1867
ecf_5R501	-0.6763	-14.4706	0.0467	0.0000	-0.7679	-0.5847
ecf_5R502	-0.5335	-12.5925	0.0424	0.0000	-0.6166	-0.4505
ecf_5R503	0.4407	4.2660	0.1033	0.0000	0.2382	0.6432
ecf_5R504	0.3421	5.2954	0.0646	0.0000	0.2155	0.4687
ecf_5R507	-0.4328	-4.5754	0.0946	0.0000	-0.6182	-0.2474
ecf_5R508	0.5356	2.7400	0.1955	0.0061	0.1524	0.9187
ecf_5R509	0.3988	3.9560	0.1008	0.0001	0.2012	0.5964
ecf_5R510	-0.3523	-4.1347	0.0852	0.0000	-0.5193	-0.1853
ecf_5R514	0.2275	3.7721	0.0603	0.0002	0.1093	0.3457
ecf_5R517	0.6276	3.7423	0.1677	0.0002	0.2989	0.9564
ecf_5R519	0.7836	2.3470	0.3339	0.0189	0.1292	1.4380
ecf_5R521	-0.2255	-2.5312	0.0891	0.0114	-0.4002	-0.0509
ecf_5R523	-0.5805	-6.3790	0.0910	0.0000	-0.7589	-0.4021

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_5R524	1.1467	3.3997	0.3373	0.0007	0.4856	1.8078
ecf_5R528	-0.6080	-8.5122	0.0714	0.0000	-0.7479	-0.4680
ecf_5R529	0.2097	3.6064	0.0582	0.0003	0.0957	0.3237
ecf_5R530	0.6909	4.8123	0.1436	0.0000	0.4095	0.9722
ecf_5R531	-0.1578	-2.1906	0.0720	0.0285	-0.2990	-0.0166
ecf_5R532	0.4090	3.3427	0.1224	0.0008	0.1692	0.6488
ecf_5R533	-0.7943	-10.8572	0.0732	0.0000	-0.9377	-0.6509
ecf_5R534	-0.4291	-6.0253	0.0712	0.0000	-0.5687	-0.2895
ecf_5R535	0.5845	2.9980	0.1949	0.0027	0.2023	0.9666
ecf_5R538	-0.6493	-5.1136	0.1270	0.0000	-0.8982	-0.4004
ecf_6R601	-1.0482	-13.5852	0.0772	0.0000	-1.1994	-0.8969
ecf_6R602	-0.8167	-15.2937	0.0534	0.0000	-0.9214	-0.7121
ecf_6R603	-0.9105	-15.6640	0.0581	0.0000	-1.0244	-0.7965
ecf_6R604	-0.4842	-10.2751	0.0471	0.0000	-0.5765	-0.3918
ecf_6R605	-0.2841	-4.5830	0.0620	0.0000	-0.4057	-0.1626
ecf_6R606	-0.4470	-5.1017	0.0876	0.0000	-0.6188	-0.2753
ecf_6R608	0.7742	5.8328	0.1327	0.0000	0.5141	1.0344
ecf_6R609	1.1179	4.7235	0.2367	0.0000	0.6540	1.5818
ecf_6R611	-0.1431	-3.2692	0.0438	0.0011	-0.2290	-0.0573
ecf_6R612	-0.2718	-3.2907	0.0826	0.0010	-0.4337	-0.1099
ecf_6R614	1.1380	8.2562	0.1378	0.0000	0.8678	1.4081
ecf_6R616	0.5062	2.8291	0.1789	0.0047	0.1555	0.8568
ecf_6R617	0.3533	1.6666	0.2120	0.0956	-0.0622	0.7688
ecf_6R618	-0.5333	-4.6159	0.1155	0.0000	-0.7598	-0.3068
ecf_6R620	-0.5242	-15.6656	0.0335	0.0000	-0.5898	-0.4586
ecf_7R701	0.1308	2.7454	0.0476	0.0060	0.0374	0.2241
ecf_7R703	-0.5508	-9.3403	0.0590	0.0000	-0.6664	-0.4352
ecf_7R704	-0.3985	-14.5623	0.0274	0.0000	-0.4521	-0.3449
ecf_7R705	-0.4020	-19.4344	0.0207	0.0000	-0.4425	-0.3614
ecf_7R706	-0.3741	-5.9278	0.0631	0.0000	-0.4977	-0.2504
ecf_7R707	-0.4921	-7.2249	0.0681	0.0000	-0.6256	-0.3586
ecf_7R708	-0.4847	-13.2508	0.0366	0.0000	-0.5564	-0.4130
ecf_7R709	-0.4606	-12.2226	0.0377	0.0000	-0.5344	-0.3867
ecf_7R710	-0.2451	-6.2539	0.0392	0.0000	-0.3219	-0.1683
ecf_7R711	-0.5300	-15.4082	0.0344	0.0000	-0.5974	-0.4626
ecf_7R712	-0.4812	-8.5491	0.0563	0.0000	-0.5916	-0.3709
ecf_7R713	-0.4296	-8.9496	0.0480	0.0000	-0.5237	-0.3355
ecf_7R714	-0.8620	-4.8199	0.1788	0.0000	-1.2125	-0.5115
ecf_7R715	-0.6680	-7.6539	0.0873	0.0000	-0.8390	-0.4969
ecf_7R716	-0.9989	-18.2508	0.0547	0.0000	-1.1062	-0.8916
ecf_7R717	-0.9433	-19.2305	0.0491	0.0000	-1.0395	-0.8472
ecf_7R718	-0.6854	-17.9229	0.0382	0.0000	-0.7603	-0.6104
ecf_7R719	-0.3199	-11.9340	0.0268	0.0000	-0.3725	-0.2674
ecf_7R720	-0.7949	-7.4558	0.1066	0.0000	-1.0039	-0.5859
ecf_7R722	-0.6913	-7.7729	0.0889	0.0000	-0.8657	-0.5170
ecf_7R723	-0.1770	-3.3190	0.0533	0.0009	-0.2815	-0.0725
ecf_7R724	-0.7457	-11.5151	0.0648	0.0000	-0.8726	-0.6188
months_1to9	-0.0087	-4.7216	0.0018	0.0000	-0.0123	-0.0051
months_10to16	0.0166	8.5161	0.0019	0.0000	0.0128	0.0204
months_17to24	0.0110	13.0720	0.0008	0.0000	0.0094	0.0127

Table 8: VIF / GVIF

term	vif	tolerance
ln_base_bldgsiz_ratio	10.432	0.096

term	vif	tolerance
smallest	3.353	0.298
months_10to16	3.274	0.305
two_plus_baths	3.056	0.327
small	2.577	0.388
yb_pre_1929	2.446	0.409
powder_room	2.355	0.425
qbelow_avg	2.094	0.478
qabove_avg	2.052	0.487
months_1to9	2.024	0.494
months_17to24	2.017	0.496
ecf_7R705	1.783	0.561
fireplace	1.756	0.570
ln_landratio	1.740	0.575
ecf_2R214	1.703	0.587
single_family1story	1.700	0.588
yb_1946_to_1960	1.624	0.616
hot_water	1.537	0.650
ecf_7R704	1.326	0.754
ecf_4R410	1.321	0.757
ecf_3R319	1.312	0.762
ecf_2R201	1.310	0.763
ecf_2R215	1.299	0.770
ecf_7R719	1.294	0.773
ecf_4R406	1.281	0.780
ecf_1R104	1.267	0.789
ecf_2R212	1.261	0.793
yb_post_1961	1.251	0.799
ecf_4R405	1.248	0.801
ecf_1R109	1.245	0.803
ecf_2R205	1.239	0.807
crawl	1.222	0.818
ecf_2R221	1.219	0.820
ecf_5R504	1.218	0.821
ecf_3R303	1.216	0.822
ecf_2R222	1.212	0.825
ecf_3R307	1.206	0.829
ecf_3R323	1.204	0.830
ecf_1R105	1.204	0.831
ecf_2R213	1.203	0.831
ecf_6R620	1.196	0.836
ecf_1R117	1.194	0.837
ecf_4R417	1.194	0.838
ecf_4R408	1.193	0.838
ecf_7R711	1.186	0.843
ecf_4R413	1.185	0.844
ln_garagespaces	1.179	0.848
ecf_6R611	1.175	0.851
ecf_3R320	1.175	0.851
ecf_4R411	1.174	0.852
ecf_2R225	1.167	0.857
wall	1.165	0.859
ecf_1R108	1.164	0.859
ecf_7R708	1.164	0.859
ecf_2R220	1.158	0.863
ecf_3R310	1.157	0.864
ecf_2R224	1.156	0.865
ecf_5R502	1.156	0.865

term	vif	tolerance
ecf_2R223	1.155	0.866
ecf_7R709	1.154	0.867
ecf_4R404	1.149	0.870
ecf_1R121	1.144	0.874
half_duplex1story	1.143	0.875
ecf_1R128	1.142	0.876
ecf_1R116	1.140	0.877
ecf_7R710	1.139	0.878
half_duplex2story	1.138	0.879
ecf_4R403	1.137	0.880
ecf_7R718	1.136	0.880
ecf_5R501	1.135	0.881
ecf_6R604	1.134	0.882
ecf_4R401	1.128	0.887
ecf_3R312	1.126	0.888
ecf_1R118	1.125	0.889
central_air	1.121	0.892
slab	1.120	0.893
ecf_1R136	1.119	0.894
ecf_1R122	1.114	0.898
ecf_7R717	1.111	0.900
ecf_1R110	1.108	0.902
ecf_3R325	1.107	0.904
ecf_5R503	1.105	0.905
ecf_1R135	1.104	0.905
ecf_4R422	1.103	0.906
fair	1.102	0.908
ecf_4R421	1.100	0.909
ecf_1R126	1.099	0.910
ecf_3R313	1.099	0.910
ecf_7R701	1.098	0.911
ecf_7R723	1.097	0.912
ecf_3R322	1.096	0.912
ecf_3R315	1.096	0.913
ecf_7R713	1.095	0.914
ecf_5R529	1.093	0.915
ecf_1R119	1.092	0.915
ecf_2R210	1.092	0.916
ecf_2R209	1.091	0.917
ecf_6R602	1.087	0.920
ecf_1R129	1.087	0.920
ecf_1R112	1.086	0.921
ecf_4R419	1.081	0.925
ecf_5R532	1.078	0.927
ecf_5R514	1.078	0.928
ecf_1R123	1.078	0.928
ecf_6R603	1.077	0.929
ecf_7R716	1.075	0.930
ecf_4R409	1.072	0.933
ecf_6R605	1.070	0.934
ecf_1R106	1.070	0.934
ecf_4R415	1.069	0.936
ecf_2R202	1.068	0.936
ecf_3R304	1.068	0.936
ecf_7R712	1.067	0.937
ecf_4R412	1.064	0.940
ecf_1R113	1.062	0.941

term	vif	tolerance
poor	1.062	0.941
ecf_7R703	1.062	0.942
ecf_4R407	1.060	0.944
ecf_3R314	1.060	0.944
ecf_1R137	1.059	0.944
ecf_5R533	1.059	0.945
ecf_4R402	1.058	0.945
ecf_1R102	1.057	0.946
ecf_3R316	1.056	0.947
ecf_7R706	1.055	0.948
ecf_7R724	1.055	0.948
ecf_5R528	1.055	0.948
ecf_3R309	1.055	0.948
good	1.054	0.948
ecf_1R124	1.054	0.949
ecf_3R321	1.053	0.949
ecf_5R509	1.052	0.951
ecf_1R120	1.051	0.951
ecf_1R127	1.051	0.951
ecf_5R531	1.050	0.953
ecf_1R114	1.049	0.953
ecf_5R534	1.049	0.953
ecf_5R507	1.047	0.955
ecf_5R510	1.045	0.957
ecf_3R311	1.044	0.958
ecf_6R601	1.044	0.958
ecf_6R612	1.043	0.958
ecf_2R208	1.043	0.958
ecf_5R523	1.043	0.959
ecf_7R707	1.043	0.959
ecf_3R324	1.042	0.960
very_poor_unsound	1.040	0.962
ecf_3R326	1.040	0.962
ecf_3R308	1.039	0.962
ecf_6R606	1.036	0.965
ecf_5R521	1.036	0.965
ecf_2R218	1.035	0.966
ecf_3R301	1.034	0.967
ecf_5R508	1.033	0.969
ecf_7R722	1.032	0.969
ecf_1R133	1.031	0.970
ecf_6R608	1.031	0.970
ecf_1R130	1.031	0.970
ecf_7R715	1.028	0.973
ecf_5R535	1.027	0.974
ecf_1R131	1.027	0.974
ecf_6R614	1.027	0.974
ecf_5R524	1.025	0.976
ecf_2R229	1.024	0.977
ecf_7R720	1.023	0.977
ecf_3R305	1.022	0.979
ecf_4R418	1.022	0.979
ecf_6R618	1.022	0.979
ecf_4R416	1.021	0.979
ecf_1R107	1.021	0.979
ecf_5R530	1.021	0.980
ecf_2R226	1.019	0.981

term	vif	tolerance
ecf_2R228	1.017	0.983
ecf_5R538	1.016	0.984
ecf_2R219	1.015	0.986
ecf_3R302	1.014	0.986
ecf_5R517	1.013	0.987
ecf_3R318	1.013	0.987
ecf_3R317	1.013	0.988
ecf_6R617	1.012	0.988
ecf_6R616	1.009	0.991
ecf_6R609	1.009	0.991
ecf_2R230	1.009	0.991
ecf_7R714	1.008	0.992
ecf_2R211	1.008	0.992
ecf_5R519	1.004	0.996

Model IV

Table 9: LN_TASP model assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
18400	1.163368	1.286896	1.092881	1.177527	37.00051	-0.5791432

Table 10: Alternative LN_TASP model assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
18399	1.165305	1.286947	1.092218	1.178288	36.94861	-0.5796658

Table 11: LN_TASP model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
18400	18215	187	0.4074	0.5678	0.5634	0.4053	0.3391	130.0333	0	19353.81	20808.35

Table 12: LN_TASP model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.5695	611.1977	0.0189	0.0000	11.5324	11.6066
ln_landratio	0.1244	7.4313	0.0167	0.0000	0.0916	0.1572
ln_base_bldgsize_ratio	0.7235	21.0273	0.0344	0.0000	0.6561	0.7910
single_family1story	0.0203	2.3301	0.0087	0.0198	0.0032	0.0374
half_duplex1story	-0.2147	-5.5867	0.0384	0.0000	-0.2901	-0.1394
half_duplex2story	-0.2980	-8.0784	0.0369	0.0000	-0.3703	-0.2257
smallest	0.0433	2.3287	0.0186	0.0199	0.0069	0.0797
small	0.0336	3.0119	0.0111	0.0026	0.0117	0.0554
two_plus_baths	0.0790	3.0633	0.0258	0.0022	0.0284	0.1295
powder_room	0.0230	1.7357	0.0133	0.0826	-0.0030	0.0490
good	0.2576	10.4201	0.0247	0.0000	0.2092	0.3061
fair	-0.1522	-19.7254	0.0077	0.0000	-0.1673	-0.1371

term	estimate	statistic	std_error	p_value	conf_low	conf_high
poor	-0.2872	-15.4638	0.0186	0.0000	-0.3236	-0.2508
very_poor_unsound	-0.2260	-5.2663	0.0429	0.0000	-0.3101	-0.1419
qabove_avg	0.1497	5.4363	0.0275	0.0000	0.0957	0.2036
qbelow_avg	-0.0595	-6.7708	0.0088	0.0000	-0.0767	-0.0423
yb_pre_1929	-0.0733	-6.7345	0.0109	0.0000	-0.0946	-0.0519
yb_1946_to_1960	0.0419	5.3167	0.0079	0.0000	0.0264	0.0573
yb_post_1961	0.0917	4.0786	0.0225	0.0000	0.0476	0.1358
crawl	-0.0864	-3.7598	0.0230	0.0002	-0.1315	-0.0414
slab	-0.1320	-4.9438	0.0267	0.0000	-0.1844	-0.0797
ln_garagespaces	0.0980	12.0027	0.0082	0.0000	0.0820	0.1140
fireplace	0.0633	7.7005	0.0082	0.0000	0.0472	0.0795
hot_water	0.0400	3.2035	0.0125	0.0014	0.0155	0.0644
wall	-0.2201	-5.0484	0.0436	0.0000	-0.3055	-0.1346
central_air	0.0506	5.9874	0.0084	0.0000	0.0340	0.0671
ecf_1R102	-0.3288	-5.3037	0.0620	0.0000	-0.4503	-0.2073
ecf_1R104	-0.1329	-4.8273	0.0275	0.0000	-0.1868	-0.0789
ecf_1R105	-0.3344	-10.7846	0.0310	0.0000	-0.3952	-0.2736
ecf_1R106	-0.2726	-5.6323	0.0484	0.0000	-0.3674	-0.1777
ecf_1R107	-0.2519	-2.5930	0.0971	0.0095	-0.4423	-0.0615
ecf_1R108	-0.4637	-14.5526	0.0319	0.0000	-0.5262	-0.4013
ecf_1R109	-0.1616	-6.0346	0.0268	0.0000	-0.2141	-0.1091
ecf_1R110	-0.1379	-3.2502	0.0424	0.0012	-0.2210	-0.0547
ecf_1R112	-0.1373	-2.7818	0.0494	0.0054	-0.2341	-0.0406
ecf_1R113	-0.5585	-9.7237	0.0574	0.0000	-0.6711	-0.4459
ecf_1R114	-0.2967	-5.0897	0.0583	0.0000	-0.4110	-0.1825
ecf_1R116	0.1294	3.5883	0.0361	0.0003	0.0587	0.2001
ecf_1R117	-0.1620	-5.2469	0.0309	0.0000	-0.2225	-0.1015
ecf_1R118	-0.2475	-6.7659	0.0366	0.0000	-0.3192	-0.1758
ecf_1R119	-0.4671	-11.4546	0.0408	0.0000	-0.5470	-0.3871
ecf_1R120	-0.1154	-2.1342	0.0541	0.0328	-0.2214	-0.0094
ecf_1R121	-0.2078	-6.2478	0.0333	0.0000	-0.2730	-0.1426
ecf_1R122	-0.4056	-11.1957	0.0362	0.0000	-0.4766	-0.3346
ecf_1R123	-0.3039	-6.2643	0.0485	0.0000	-0.3990	-0.2088
ecf_1R124	-0.6375	-8.6050	0.0741	0.0000	-0.7827	-0.4923
ecf_1R125	-0.1191	-1.8873	0.0631	0.0591	-0.2428	0.0046
ecf_1R126	0.2244	5.6885	0.0394	0.0000	0.1471	0.3017
ecf_1R127	-0.4047	-7.6007	0.0532	0.0000	-0.5091	-0.3003
ecf_1R128	-0.5356	-15.5146	0.0345	0.0000	-0.6032	-0.4679
ecf_1R129	-0.1841	-4.1419	0.0444	0.0000	-0.2712	-0.0970
ecf_1R130	NA	NA	NA	NA	NA	NA
ecf_1R131	-0.3796	-3.5627	0.1066	0.0004	-0.5885	-0.1708
ecf_1R133	-0.4314	-5.3104	0.0812	0.0000	-0.5906	-0.2721
ecf_1R135	-0.2530	-6.4862	0.0390	0.0000	-0.3294	-0.1765
ecf_1R136	-0.3121	-8.8720	0.0352	0.0000	-0.3810	-0.2431
ecf_1R137	-0.7734	-13.2370	0.0584	0.0000	-0.8879	-0.6589
ecf_2R201	-0.3146	-12.0875	0.0260	0.0000	-0.3657	-0.2636
ecf_2R202	-0.1471	-3.1869	0.0462	0.0014	-0.2376	-0.0566
ecf_2R203	-0.0496	-1.7894	0.0277	0.0736	-0.1039	0.0047
ecf_2R205	-0.2133	-7.2043	0.0296	0.0000	-0.2714	-0.1553
ecf_2R208	0.4768	7.3749	0.0647	0.0000	0.3501	0.6035
ecf_2R209	0.3335	3.2337	0.1031	0.0012	0.1313	0.5356
ecf_2R210	0.3957	2.7990	0.1414	0.0051	0.1186	0.6729
ecf_2R211	-1.0890	-3.7656	0.2892	0.0002	-1.6558	-0.5221
ecf_2R212	-0.1971	-7.2756	0.0271	0.0000	-0.2501	-0.1440
ecf_2R213	-0.1070	-3.5710	0.0300	0.0004	-0.1657	-0.0483
ecf_2R214	0.0508	2.4505	0.0207	0.0143	0.0102	0.0915
ecf_2R215	0.3010	5.6738	0.0530	0.0000	0.1970	0.4049

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_2R216	0.2693	1.7056	0.1579	0.0881	-0.0402	0.5789
ecf_2R217	-0.4043	-1.7078	0.2368	0.0877	-0.8684	0.0597
ecf_2R218	-0.5972	-6.9257	0.0862	0.0000	-0.7662	-0.4282
ecf_2R219	0.2677	1.8457	0.1450	0.0650	-0.0166	0.5519
ecf_2R220	-0.2688	-8.0347	0.0335	0.0000	-0.3344	-0.2032
ecf_2R221	-0.2088	-6.9949	0.0299	0.0000	-0.2673	-0.1503
ecf_2R222	-0.3740	-12.3317	0.0303	0.0000	-0.4335	-0.3146
ecf_2R223	-0.2860	-8.6079	0.0332	0.0000	-0.3511	-0.2209
ecf_2R224	-0.4824	-13.6434	0.0354	0.0000	-0.5517	-0.4131
ecf_2R225	-0.3084	-8.8693	0.0348	0.0000	-0.3766	-0.2403
ecf_2R226	-0.6645	-5.1148	0.1299	0.0000	-0.9191	-0.4098
ecf_2R228	-0.9840	-6.3471	0.1550	0.0000	-1.2879	-0.6801
ecf_2R229	-0.7189	-5.2585	0.1367	0.0000	-0.9869	-0.4510
ecf_2R230	-0.9088	-2.2262	0.4082	0.0260	-1.7090	-0.1086
ecf_3R301	-0.5928	-4.9935	0.1187	0.0000	-0.8255	-0.3601
ecf_3R303	-0.6149	-21.0449	0.0292	0.0000	-0.6722	-0.5577
ecf_3R304	-0.8293	-12.9339	0.0641	0.0000	-0.9550	-0.7036
ecf_3R305	-1.0313	-6.1630	0.1673	0.0000	-1.3593	-0.7033
ecf_3R306	0.1667	3.1668	0.0526	0.0015	0.0635	0.2699
ecf_3R307	-0.4837	-16.4147	0.0295	0.0000	-0.5415	-0.4259
ecf_3R308	-0.8127	-10.8852	0.0747	0.0000	-0.9590	-0.6663
ecf_3R309	-0.8281	-10.6999	0.0774	0.0000	-0.9798	-0.6764
ecf_3R310	0.3539	8.7688	0.0404	0.0000	0.2748	0.4330
ecf_3R311	-0.5290	-7.0697	0.0748	0.0000	-0.6756	-0.3823
ecf_3R312	-0.7647	-20.6527	0.0370	0.0000	-0.8373	-0.6922
ecf_3R313	-0.7017	-16.6458	0.0422	0.0000	-0.7843	-0.6190
ecf_3R314	-0.3547	-6.6501	0.0533	0.0000	-0.4592	-0.2501
ecf_3R315	-0.3931	-9.6166	0.0409	0.0000	-0.4732	-0.3130
ecf_3R316	-0.6207	-10.3512	0.0600	0.0000	-0.7383	-0.5032
ecf_3R317	-0.9743	-5.3266	0.1829	0.0000	-1.3328	-0.6158
ecf_3R318	-0.3444	-2.9025	0.1187	0.0037	-0.5770	-0.1118
ecf_3R319	-0.6355	-24.7268	0.0257	0.0000	-0.6858	-0.5851
ecf_3R320	-0.5428	-16.9510	0.0320	0.0000	-0.6056	-0.4801
ecf_3R321	-0.8384	-14.6368	0.0573	0.0000	-0.9506	-0.7261
ecf_3R322	-0.2984	-7.2314	0.0413	0.0000	-0.3793	-0.2175
ecf_3R323	-0.4819	-16.0738	0.0300	0.0000	-0.5406	-0.4231
ecf_3R324	-0.7965	-11.9126	0.0669	0.0000	-0.9275	-0.6654
ecf_3R325	-0.5541	-12.9469	0.0428	0.0000	-0.6380	-0.4702
ecf_3R326	-1.2016	-15.7805	0.0761	0.0000	-1.3508	-1.0523
ecf_4R401	-0.6636	-18.9930	0.0349	0.0000	-0.7321	-0.5951
ecf_4R402	-1.0661	-18.2282	0.0585	0.0000	-1.1808	-0.9515
ecf_4R403	-0.5422	-13.8053	0.0393	0.0000	-0.6192	-0.4652
ecf_4R404	-0.4931	-12.5574	0.0393	0.0000	-0.5701	-0.4162
ecf_4R405	-0.3992	-14.5170	0.0275	0.0000	-0.4531	-0.3453
ecf_4R406	-0.4688	-17.7467	0.0264	0.0000	-0.5206	-0.4170
ecf_4R407	-0.5862	-11.4073	0.0514	0.0000	-0.6869	-0.4855
ecf_4R408	-0.5919	-19.2643	0.0307	0.0000	-0.6521	-0.5316
ecf_4R409	-0.4826	-10.6512	0.0453	0.0000	-0.5714	-0.3938
ecf_4R410	-0.1931	-7.8448	0.0246	0.0000	-0.2414	-0.1449
ecf_4R411	-0.6614	-21.5205	0.0307	0.0000	-0.7217	-0.6012
ecf_4R412	-0.7557	-14.1582	0.0534	0.0000	-0.8604	-0.6511
ecf_4R413	-0.3131	-10.1916	0.0307	0.0000	-0.3733	-0.2529
ecf_4R415	0.1721	3.6633	0.0470	0.0002	0.0800	0.2641
ecf_4R416	-0.6861	-6.4508	0.1064	0.0000	-0.8945	-0.4776
ecf_4R417	-0.1699	-5.6110	0.0303	0.0000	-0.2293	-0.1106
ecf_4R418	-0.5873	-4.5190	0.1300	0.0000	-0.8420	-0.3326
ecf_4R419	-0.8183	-14.3795	0.0569	0.0000	-0.9299	-0.7068

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_4R421	-0.7425	-10.8819	0.0682	0.0000	-0.8763	-0.6088
ecf_4R422	0.1140	2.7461	0.0415	0.0060	0.0326	0.1955
ecf_5R501	-0.7884	-17.1372	0.0460	0.0000	-0.8786	-0.6982
ecf_5R502	-0.6737	-16.0949	0.0419	0.0000	-0.7557	-0.5916
ecf_5R503	0.4793	4.8774	0.0983	0.0000	0.2867	0.6719
ecf_5R504	0.3806	6.5653	0.0580	0.0000	0.2669	0.4942
ecf_5R505	0.4146	2.0189	0.2054	0.0435	0.0121	0.8171
ecf_5R507	-0.5009	-4.5274	0.1106	0.0000	-0.7177	-0.2840
ecf_5R508	0.6806	1.6642	0.4090	0.0961	-0.1210	1.4822
ecf_5R509	0.4167	3.7460	0.1112	0.0002	0.1987	0.6347
ecf_5R510	-0.7541	-5.5055	0.1370	0.0000	-1.0226	-0.4857
ecf_5R514	0.2118	3.4659	0.0611	0.0005	0.0920	0.3316
ecf_5R517	0.5891	3.2136	0.1833	0.0013	0.2298	0.9484
ecf_5R519	0.7442	2.5771	0.2888	0.0100	0.1782	1.3102
ecf_5R521	-0.3768	-2.5960	0.1451	0.0094	-0.6612	-0.0923
ecf_5R523	-0.6378	-4.3859	0.1454	0.0000	-0.9228	-0.3527
ecf_5R524	NA	NA	NA	NA	NA	NA
ecf_5R528	-0.6947	-9.3953	0.0739	0.0000	-0.8396	-0.5498
ecf_5R529	0.2641	4.3829	0.0603	0.0000	0.1460	0.3822
ecf_5R530	0.6571	2.2724	0.2892	0.0231	0.0903	1.2239
ecf_5R531	-0.2411	-2.5424	0.0948	0.0110	-0.4270	-0.0552
ecf_5R532	0.4623	3.3052	0.1399	0.0010	0.1881	0.7364
ecf_5R533	-0.8555	-10.2591	0.0834	0.0000	-1.0190	-0.6921
ecf_5R534	-0.5298	-6.1226	0.0865	0.0000	-0.6994	-0.3602
ecf_5R538	-0.8110	-4.4331	0.1829	0.0000	-1.1695	-0.4524
ecf_5R539	-0.3083	-1.9884	0.1550	0.0468	-0.6122	-0.0044
ecf_6R601	-1.1281	-14.0908	0.0801	0.0000	-1.2851	-0.9712
ecf_6R602	-0.8790	-16.3368	0.0538	0.0000	-0.9844	-0.7735
ecf_6R603	-1.0497	-16.6693	0.0630	0.0000	-1.1731	-0.9263
ecf_6R604	-0.5165	-11.9417	0.0432	0.0000	-0.6012	-0.4317
ecf_6R605	-0.2864	-4.7268	0.0606	0.0000	-0.4051	-0.1676
ecf_6R606	-0.5069	-4.5985	0.1102	0.0000	-0.7230	-0.2908
ecf_6R608	0.8509	6.1692	0.1379	0.0000	0.5805	1.1212
ecf_6R609	1.0886	2.6695	0.4078	0.0076	0.2893	1.8880
ecf_6R611	-0.1566	-3.9593	0.0396	0.0001	-0.2342	-0.0791
ecf_6R612	-0.2823	-3.2585	0.0866	0.0011	-0.4520	-0.1125
ecf_6R614	1.0022	5.4497	0.1839	0.0000	0.6417	1.3626
ecf_6R618	-0.5894	-3.7994	0.1551	0.0001	-0.8935	-0.2853
ecf_6R620	-0.5306	-17.3943	0.0305	0.0000	-0.5904	-0.4708
ecf_7R701	0.1211	2.8725	0.0421	0.0041	0.0385	0.2037
ecf_7R703	-0.5899	-11.2728	0.0523	0.0000	-0.6925	-0.4874
ecf_7R704	-0.4207	-16.7949	0.0251	0.0000	-0.4698	-0.3716
ecf_7R705	-0.4102	-20.8399	0.0197	0.0000	-0.4488	-0.3716
ecf_7R706	-0.4549	-7.8524	0.0579	0.0000	-0.5685	-0.3414
ecf_7R707	-0.5029	-7.7003	0.0653	0.0000	-0.6309	-0.3749
ecf_7R708	-0.5165	-15.6487	0.0330	0.0000	-0.5812	-0.4518
ecf_7R709	-0.4706	-13.8882	0.0339	0.0000	-0.5371	-0.4042
ecf_7R710	-0.2684	-7.6464	0.0351	0.0000	-0.3372	-0.1996
ecf_7R711	-0.5455	-17.5922	0.0310	0.0000	-0.6063	-0.4847
ecf_7R712	-0.4489	-8.8584	0.0507	0.0000	-0.5483	-0.3496
ecf_7R713	-0.4568	-10.5983	0.0431	0.0000	-0.5413	-0.3723
ecf_7R714	-0.9384	-2.3008	0.4079	0.0214	-1.7378	-0.1389
ecf_7R715	-0.6383	-6.0056	0.1063	0.0000	-0.8467	-0.4300
ecf_7R716	-0.9558	-17.5479	0.0545	0.0000	-1.0625	-0.8490
ecf_7R717	-1.0384	-22.5990	0.0459	0.0000	-1.1285	-0.9484
ecf_7R718	-0.7148	-20.5209	0.0348	0.0000	-0.7831	-0.6465
ecf_7R719	-0.3382	-13.7008	0.0247	0.0000	-0.3866	-0.2898

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_7R720	-0.8272	-6.9672	0.1187	0.0000	-1.0599	-0.5945
ecf_7R722	-0.7761	-6.5326	0.1188	0.0000	-1.0089	-0.5432
ecf_7R723	-0.1395	-2.7877	0.0501	0.0053	-0.2376	-0.0414
ecf_7R724	-0.7557	-10.7080	0.0706	0.0000	-0.8941	-0.6174

Table 13: Alternative LN_TASP model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
18399	18212	189	0.4075	0.5674	0.563	0.4055	0.3392	128.4361	0	19371.97	20842.14

Table 14: Alternative LN_TASP model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	9.0726	51.7200	0.1754	0.0000	8.7288	9.4165
ln_landratio	0.1374	8.2553	0.0166	0.0000	0.1048	0.1701
lnsbldsfint	0.3872	14.5527	0.0266	0.0000	0.3351	0.4394
single_family1story	0.0234	2.7168	0.0086	0.0066	0.0065	0.0402
half_duplex2story	-0.1256	-3.4136	0.0368	0.0006	-0.1977	-0.0535
smallest	0.0554	2.3649	0.0234	0.0180	0.0095	0.1013
small	0.0247	2.0811	0.0119	0.0374	0.0014	0.0479
large	0.0313	2.5034	0.0125	0.0123	0.0068	0.0559
largest	0.0729	3.2343	0.0225	0.0012	0.0287	0.1171
two_plus_baths	0.1414	5.4258	0.0261	0.0000	0.0903	0.1925
powder_room	0.0371	2.7036	0.0137	0.0069	0.0102	0.0639
good	0.2599	10.5052	0.0247	0.0000	0.2114	0.3083
fair	-0.1517	-19.6448	0.0077	0.0000	-0.1668	-0.1366
poor	-0.2872	-15.4471	0.0186	0.0000	-0.3236	-0.2507
very_poor_unsound	-0.2289	-5.3316	0.0429	0.0000	-0.3131	-0.1448
qabove_avg	0.1468	5.2390	0.0280	0.0000	0.0919	0.2017
qbelow_avg	-0.0591	-6.7172	0.0088	0.0000	-0.0764	-0.0419
yb_pre_1929	-0.0692	-6.3500	0.0109	0.0000	-0.0905	-0.0478
yb_1946_to_1960	0.0441	5.5866	0.0079	0.0000	0.0286	0.0595
yb_post_1961	0.0950	4.2202	0.0225	0.0000	0.0509	0.1392
crawl	-0.0793	-3.4449	0.0230	0.0006	-0.1245	-0.0342
slab	-0.1270	-4.7551	0.0267	0.0000	-0.1794	-0.0747
garage	0.0789	11.3210	0.0070	0.0000	0.0652	0.0925
fireplace	0.0629	7.6437	0.0082	0.0000	0.0468	0.0790
hot_water	0.0372	2.9738	0.0125	0.0029	0.0127	0.0618
wall	-0.2155	-4.9375	0.0437	0.0000	-0.3011	-0.1300
central_air	0.0517	6.1258	0.0084	0.0000	0.0352	0.0683
ecf_1R102	-0.3332	-5.3723	0.0620	0.0000	-0.4547	-0.2116
ecf_1R104	-0.1302	-4.7275	0.0275	0.0000	-0.1842	-0.0762
ecf_1R105	-0.3341	-10.7681	0.0310	0.0000	-0.3949	-0.2733
ecf_1R106	-0.2732	-5.6416	0.0484	0.0000	-0.3681	-0.1783
ecf_1R107	-0.2502	-2.5745	0.0972	0.0100	-0.4407	-0.0597
ecf_1R108	-0.4611	-14.4558	0.0319	0.0000	-0.5236	-0.3985
ecf_1R109	-0.1604	-5.9849	0.0268	0.0000	-0.2130	-0.1079
ecf_1R110	-0.1410	-3.3232	0.0424	0.0009	-0.2242	-0.0578
ecf_1R112	-0.1321	-2.6734	0.0494	0.0075	-0.2289	-0.0352
ecf_1R113	-0.5559	-9.6723	0.0575	0.0000	-0.6685	-0.4432
ecf_1R114	-0.2957	-5.0680	0.0583	0.0000	-0.4100	-0.1813
ecf_1R116	0.1330	3.6805	0.0361	0.0002	0.0622	0.2039
ecf_1R117	-0.1615	-5.2228	0.0309	0.0000	-0.2221	-0.1009

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_1R118	-0.2477	-6.7662	0.0366	0.0000	-0.3194	-0.1759
ecf_1R119	-0.4664	-11.4275	0.0408	0.0000	-0.5464	-0.3864
ecf_1R120	-0.1213	-2.2407	0.0541	0.0251	-0.2273	-0.0152
ecf_1R121	-0.2070	-6.2175	0.0333	0.0000	-0.2723	-0.1418
ecf_1R122	-0.4036	-11.1296	0.0363	0.0000	-0.4746	-0.3325
ecf_1R123	-0.3006	-6.1934	0.0485	0.0000	-0.3958	-0.2055
ecf_1R124	-0.6356	-8.5744	0.0741	0.0000	-0.7809	-0.4903
ecf_1R125	-0.1234	-1.9550	0.0631	0.0506	-0.2471	0.0003
ecf_1R126	0.2214	5.6042	0.0395	0.0000	0.1439	0.2988
ecf_1R127	-0.4016	-7.5352	0.0533	0.0000	-0.5061	-0.2971
ecf_1R128	-0.5352	-15.4826	0.0346	0.0000	-0.6030	-0.4675
ecf_1R129	-0.1842	-4.1406	0.0445	0.0000	-0.2713	-0.0970
ecf_1R130	NA	NA	NA	NA	NA	NA
ecf_1R131	-0.3740	-3.5080	0.1066	0.0005	-0.5830	-0.1650
ecf_1R133	-0.4259	-5.2399	0.0813	0.0000	-0.5852	-0.2666
ecf_1R135	-0.2540	-6.5044	0.0390	0.0000	-0.3305	-0.1774
ecf_1R136	-0.3138	-8.9152	0.0352	0.0000	-0.3828	-0.2448
ecf_1R137	-0.7708	-13.1824	0.0585	0.0000	-0.8854	-0.6562
ecf_2R201	-0.3127	-12.0013	0.0261	0.0000	-0.3638	-0.2616
ecf_2R202	-0.1489	-3.2226	0.0462	0.0013	-0.2394	-0.0583
ecf_2R203	-0.0491	-1.7673	0.0278	0.0772	-0.1035	0.0054
ecf_2R205	-0.2098	-7.0779	0.0296	0.0000	-0.2679	-0.1517
ecf_2R208	0.4745	7.3308	0.0647	0.0000	0.3477	0.6014
ecf_2R209	0.3828	3.7015	0.1034	0.0002	0.1801	0.5855
ecf_2R210	0.6143	4.3424	0.1415	0.0000	0.3370	0.8917
ecf_2R211	-1.0821	-3.7397	0.2894	0.0002	-1.6493	-0.5149
ecf_2R212	-0.1985	-7.3191	0.0271	0.0000	-0.2516	-0.1453
ecf_2R213	-0.1063	-3.5454	0.0300	0.0004	-0.1651	-0.0475
ecf_2R214	0.0505	2.4201	0.0209	0.0155	0.0096	0.0915
ecf_2R215	0.3199	5.9723	0.0536	0.0000	0.2149	0.4248
ecf_2R216	0.3867	2.4450	0.1582	0.0145	0.0767	0.6967
ecf_2R218	-0.5901	-6.8380	0.0863	0.0000	-0.7593	-0.4210
ecf_2R219	0.2614	1.8009	0.1452	0.0717	-0.0231	0.5459
ecf_2R220	-0.2670	-7.9695	0.0335	0.0000	-0.3327	-0.2014
ecf_2R221	-0.2067	-6.9074	0.0299	0.0000	-0.2654	-0.1480
ecf_2R222	-0.3724	-12.2622	0.0304	0.0000	-0.4319	-0.3128
ecf_2R223	-0.2847	-8.5566	0.0333	0.0000	-0.3499	-0.2195
ecf_2R224	-0.4808	-13.5753	0.0354	0.0000	-0.5502	-0.4114
ecf_2R225	-0.3068	-8.8080	0.0348	0.0000	-0.3751	-0.2385
ecf_2R226	-0.6646	-5.1201	0.1298	0.0000	-0.9190	-0.4102
ecf_2R228	-0.9824	-6.3328	0.1551	0.0000	-1.2864	-0.6783
ecf_2R229	-0.7062	-5.1614	0.1368	0.0000	-0.9743	-0.4380
ecf_2R230	-0.9324	-2.2828	0.4085	0.0225	-1.7330	-0.1318
ecf_3R301	-0.5879	-4.9494	0.1188	0.0000	-0.8207	-0.3551
ecf_3R303	-0.6072	-20.7547	0.0293	0.0000	-0.6646	-0.5499
ecf_3R304	-0.8271	-12.8910	0.0642	0.0000	-0.9529	-0.7013
ecf_3R305	-1.0253	-6.1229	0.1674	0.0000	-1.3535	-0.6971
ecf_3R306	0.1694	3.2151	0.0527	0.0013	0.0661	0.2727
ecf_3R307	-0.4805	-16.2895	0.0295	0.0000	-0.5383	-0.4227
ecf_3R308	-0.8033	-10.7549	0.0747	0.0000	-0.9497	-0.6569
ecf_3R309	-0.8259	-10.6642	0.0774	0.0000	-0.9777	-0.6741
ecf_3R310	0.3565	8.7922	0.0405	0.0000	0.2770	0.4360
ecf_3R311	-0.5264	-7.0319	0.0749	0.0000	-0.6732	-0.3797
ecf_3R312	-0.7628	-20.5821	0.0371	0.0000	-0.8354	-0.6901
ecf_3R313	-0.6928	-16.4111	0.0422	0.0000	-0.7756	-0.6101
ecf_3R314	-0.3496	-6.5510	0.0534	0.0000	-0.4542	-0.2450
ecf_3R315	-0.3902	-9.5363	0.0409	0.0000	-0.4704	-0.3100

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_3R316	-0.6155	-10.2571	0.0600	0.0000	-0.7331	-0.4979
ecf_3R317	-0.9753	-5.3288	0.1830	0.0000	-1.3340	-0.6165
ecf_3R318	-0.3451	-2.9066	0.1187	0.0037	-0.5778	-0.1124
ecf_3R319	-0.6332	-24.6096	0.0257	0.0000	-0.6836	-0.5828
ecf_3R320	-0.5385	-16.8067	0.0320	0.0000	-0.6014	-0.4757
ecf_3R321	-0.8367	-14.5961	0.0573	0.0000	-0.9491	-0.7243
ecf_3R322	-0.2989	-7.2381	0.0413	0.0000	-0.3799	-0.2180
ecf_3R323	-0.4794	-15.9787	0.0300	0.0000	-0.5382	-0.4206
ecf_3R324	-0.7925	-11.8460	0.0669	0.0000	-0.9236	-0.6613
ecf_3R325	-0.5499	-12.8416	0.0428	0.0000	-0.6338	-0.4659
ecf_3R326	-1.2007	-15.7578	0.0762	0.0000	-1.3501	-1.0514
ecf_4R401	-0.6636	-18.9720	0.0350	0.0000	-0.7321	-0.5950
ecf_4R402	-1.0676	-18.2407	0.0585	0.0000	-1.1823	-0.9529
ecf_4R403	-0.5461	-13.9468	0.0392	0.0000	-0.6229	-0.4694
ecf_4R404	-0.4969	-12.7416	0.0390	0.0000	-0.5733	-0.4204
ecf_4R405	-0.4028	-14.6351	0.0275	0.0000	-0.4568	-0.3489
ecf_4R406	-0.4690	-17.7226	0.0265	0.0000	-0.5209	-0.4171
ecf_4R407	-0.5879	-11.4291	0.0514	0.0000	-0.6887	-0.4871
ecf_4R408	-0.5899	-19.1668	0.0308	0.0000	-0.6502	-0.5296
ecf_4R409	-0.4824	-10.6377	0.0453	0.0000	-0.5713	-0.3935
ecf_4R410	-0.1901	-7.7124	0.0247	0.0000	-0.2385	-0.1418
ecf_4R411	-0.6609	-21.4764	0.0308	0.0000	-0.7212	-0.6005
ecf_4R412	-0.7549	-14.1392	0.0534	0.0000	-0.8596	-0.6503
ecf_4R413	-0.3132	-10.1797	0.0308	0.0000	-0.3735	-0.2529
ecf_4R415	0.1725	3.6668	0.0470	0.0002	0.0803	0.2647
ecf_4R416	-0.6863	-6.4490	0.1064	0.0000	-0.8948	-0.4777
ecf_4R417	-0.1707	-5.6264	0.0303	0.0000	-0.2302	-0.1113
ecf_4R418	-0.5902	-4.5390	0.1300	0.0000	-0.8451	-0.3353
ecf_4R419	-0.8244	-14.4724	0.0570	0.0000	-0.9361	-0.7128
ecf_4R421	-0.7492	-10.9658	0.0683	0.0000	-0.8831	-0.6153
ecf_4R422	0.1160	2.7887	0.0416	0.0053	0.0345	0.1975
ecf_5R501	-0.7827	-16.9838	0.0461	0.0000	-0.8730	-0.6923
ecf_5R502	-0.6703	-15.9902	0.0419	0.0000	-0.7525	-0.5881
ecf_5R503	0.5580	5.6554	0.0987	0.0000	0.3646	0.7514
ecf_5R504	0.3887	6.6568	0.0584	0.0000	0.2742	0.5031
ecf_5R505	0.4061	1.9754	0.2056	0.0482	0.0032	0.8090
ecf_5R506	0.2034	1.6706	0.1217	0.0948	-0.0352	0.4419
ecf_5R507	-0.4984	-4.4994	0.1108	0.0000	-0.7155	-0.2813
ecf_5R508	0.8153	1.9918	0.4093	0.0464	0.0130	1.6176
ecf_5R509	0.4432	3.9778	0.1114	0.0001	0.2248	0.6617
ecf_5R510	-0.7531	-5.4962	0.1370	0.0000	-1.0217	-0.4845
ecf_5R514	0.2184	3.5653	0.0613	0.0004	0.0983	0.3385
ecf_5R517	0.5983	3.2621	0.1834	0.0011	0.2388	0.9578
ecf_5R519	0.7318	2.5326	0.2889	0.0113	0.1654	1.2981
ecf_5R521	-0.3721	-2.5618	0.1452	0.0104	-0.6568	-0.0874
ecf_5R523	-0.6379	-4.3843	0.1455	0.0000	-0.9231	-0.3527
ecf_5R524	NA	NA	NA	NA	NA	NA
ecf_5R528	-0.6886	-9.3050	0.0740	0.0000	-0.8336	-0.5435
ecf_5R529	0.2722	4.5113	0.0603	0.0000	0.1539	0.3905
ecf_5R530	0.6516	2.2521	0.2893	0.0243	0.0845	1.2186
ecf_5R531	-0.2331	-2.4560	0.0949	0.0141	-0.4192	-0.0471
ecf_5R532	0.6088	4.3437	0.1402	0.0000	0.3341	0.8835
ecf_5R533	-0.8603	-10.3049	0.0835	0.0000	-1.0239	-0.6966
ecf_5R534	-0.5333	-6.1580	0.0866	0.0000	-0.7030	-0.3635
ecf_5R535	0.5101	1.7579	0.2902	0.0788	-0.0587	1.0788
ecf_5R538	-0.8215	-4.4873	0.1831	0.0000	-1.1803	-0.4626
ecf_5R539	-0.3163	-2.0391	0.1551	0.0415	-0.6204	-0.0123

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_6R601	-1.1293	-14.0967	0.0801	0.0000	-1.2864	-0.9723
ecf_6R602	-0.8752	-16.2538	0.0538	0.0000	-0.9807	-0.7697
ecf_6R603	-1.0477	-16.6234	0.0630	0.0000	-1.1713	-0.9242
ecf_6R604	-0.5050	-11.6632	0.0433	0.0000	-0.5899	-0.4201
ecf_6R605	-0.2751	-4.5377	0.0606	0.0000	-0.3940	-0.1563
ecf_6R606	-0.4946	-4.4846	0.1103	0.0000	-0.7108	-0.2784
ecf_6R608	0.8560	6.2018	0.1380	0.0000	0.5855	1.1265
ecf_6R609	1.0634	2.6057	0.4081	0.0092	0.2635	1.8633
ecf_6R611	-0.1525	-3.8490	0.0396	0.0001	-0.2302	-0.0748
ecf_6R612	-0.2582	-2.9780	0.0867	0.0029	-0.4281	-0.0882
ecf_6R614	1.0075	5.4757	0.1840	0.0000	0.6469	1.3682
ecf_6R618	-0.5744	-3.7004	0.1552	0.0002	-0.8786	-0.2701
ecf_6R620	-0.5231	-17.1340	0.0305	0.0000	-0.5829	-0.4632
ecf_7R701	0.1208	2.8628	0.0422	0.0042	0.0381	0.2034
ecf_7R703	-0.5876	-11.2228	0.0524	0.0000	-0.6902	-0.4850
ecf_7R704	-0.4216	-16.8141	0.0251	0.0000	-0.4707	-0.3724
ecf_7R705	-0.4054	-20.5740	0.0197	0.0000	-0.4440	-0.3668
ecf_7R706	-0.4555	-7.8575	0.0580	0.0000	-0.5692	-0.3419
ecf_7R707	-0.4951	-7.5760	0.0653	0.0000	-0.6232	-0.3670
ecf_7R708	-0.5149	-15.5841	0.0330	0.0000	-0.5796	-0.4501
ecf_7R709	-0.4696	-13.8481	0.0339	0.0000	-0.5361	-0.4032
ecf_7R710	-0.2655	-7.5581	0.0351	0.0000	-0.3343	-0.1966
ecf_7R711	-0.5437	-17.5143	0.0310	0.0000	-0.6046	-0.4829
ecf_7R712	-0.4485	-8.8438	0.0507	0.0000	-0.5479	-0.3491
ecf_7R713	-0.4551	-10.5499	0.0431	0.0000	-0.5396	-0.3705
ecf_7R714	-0.9276	-2.2729	0.4081	0.0230	-1.7276	-0.1277
ecf_7R715	-0.6373	-5.9928	0.1063	0.0000	-0.8457	-0.4289
ecf_7R716	-0.9558	-17.5284	0.0545	0.0000	-1.0627	-0.8489
ecf_7R717	-1.0432	-22.6696	0.0460	0.0000	-1.1334	-0.9530
ecf_7R718	-0.7120	-20.4154	0.0349	0.0000	-0.7803	-0.6436
ecf_7R719	-0.3372	-13.6452	0.0247	0.0000	-0.3857	-0.2888
ecf_7R720	-0.8255	-6.9471	0.1188	0.0000	-1.0584	-0.5926
ecf_7R722	-0.7744	-6.5146	0.1189	0.0000	-1.0073	-0.5414
ecf_7R723	-0.1473	-2.9287	0.0503	0.0034	-0.2458	-0.0487
ecf_7R724	-0.7524	-10.6532	0.0706	0.0000	-0.8909	-0.6140

Table 15: VIF / GVIF

term	vif	tolerance
ln_base_bldgsize_ratio	10.606	0.094
smallest	3.411	0.293
two_plus_baths	3.016	0.332
small	2.619	0.382
powder_room	2.399	0.417
yb_pre_1929	2.330	0.429
ecf_7R705	2.114	0.473
qbelow_avg	2.104	0.475
qabove_avg	2.057	0.486
ecf_2R214	2.042	0.490
fireplace	1.775	0.563
ln_landratio	1.749	0.572
single_family1story	1.624	0.616
yb_1946_to_1960	1.603	0.624
hot_water	1.533	0.652
ecf_7R704	1.473	0.679

term	vif	tolerance
ecf_4R410	1.468	0.681
ecf_2R201	1.453	0.688
ecf_7R719	1.451	0.689
ecf_3R319	1.451	0.689
ecf_4R406	1.416	0.706
ecf_2R212	1.386	0.722
ecf_1R104	1.382	0.723
ecf_1R109	1.371	0.729
ecf_2R215	1.367	0.731
ecf_4R405	1.358	0.736
ecf_2R203	1.344	0.744
ecf_2R205	1.334	0.750
ecf_2R221	1.324	0.755
ecf_2R222	1.313	0.761
ecf_3R303	1.309	0.764
ecf_3R307	1.306	0.766
ecf_2R213	1.303	0.768
ecf_3R323	1.299	0.770
ecf_1R117	1.288	0.776
ecf_1R105	1.288	0.776
ecf_4R408	1.286	0.777
ecf_4R417	1.282	0.780
ecf_6R620	1.279	0.782
ecf_7R711	1.277	0.783
ecf_4R413	1.275	0.784
ecf_3R320	1.253	0.798
ecf_5R504	1.251	0.799
ecf_4R411	1.249	0.801
yb_post_1961	1.242	0.805
ecf_6R611	1.236	0.809
ecf_2R220	1.235	0.809
ecf_1R108	1.234	0.810
ecf_7R708	1.234	0.810
ecf_2R223	1.231	0.812
crawl	1.231	0.813
ecf_2R225	1.227	0.815
ecf_2R224	1.225	0.817
ecf_7R709	1.220	0.820
ecf_1R121	1.215	0.823
ecf_1R128	1.210	0.827
ecf_4R404	1.200	0.834
ecf_3R310	1.199	0.834
ecf_7R710	1.199	0.834
ecf_1R116	1.197	0.835
ecf_7R718	1.196	0.836
ecf_4R401	1.189	0.841
ecf_1R118	1.184	0.845
ecf_1R136	1.183	0.846
ecf_4R403	1.182	0.846
ecf_3R312	1.181	0.847
ecf_6R604	1.177	0.850
ecf_5R502	1.175	0.851
ecf_1R122	1.169	0.856
ln_garagespaces	1.169	0.856
ecf_1R135	1.156	0.865
wall	1.150	0.869
ecf_1R126	1.145	0.873

term	vif	tolerance
ecf_1R110	1.143	0.875
ecf_5R501	1.142	0.875
ecf_3R322	1.142	0.875
ecf_3R325	1.142	0.876
ecf_3R315	1.141	0.877
ecf_7R717	1.140	0.878
ecf_3R313	1.139	0.878
ecf_7R701	1.139	0.878
ecf_4R422	1.137	0.880
ecf_7R713	1.136	0.881
ecf_1R119	1.135	0.881
half_duplex1story	1.132	0.884
ecf_3R306	1.132	0.884
half_duplex2story	1.131	0.884
ecf_1R129	1.125	0.889
ecf_4R421	1.120	0.893
central_air	1.116	0.896
ecf_7R723	1.113	0.899
ecf_1R112	1.112	0.900
ecf_4R409	1.108	0.902
ecf_5R503	1.104	0.905
ecf_1R123	1.102	0.908
slab	1.101	0.908
ecf_2R202	1.099	0.910
ecf_4R415	1.099	0.910
ecf_1R106	1.096	0.912
ecf_6R602	1.095	0.913
ecf_7R712	1.095	0.913
ecf_5R529	1.091	0.917
ecf_4R419	1.090	0.918
ecf_2R209	1.089	0.919
fair	1.087	0.920
ecf_7R716	1.087	0.920
ecf_7R703	1.085	0.921
ecf_2R210	1.084	0.923
ecf_6R605	1.081	0.925
ecf_4R407	1.078	0.928
ecf_4R412	1.078	0.928
ecf_5R514	1.078	0.928
ecf_3R314	1.076	0.929
ecf_6R603	1.073	0.932
ecf_1R127	1.073	0.932
ecf_1R120	1.071	0.934
ecf_1R113	1.070	0.934
ecf_7R706	1.069	0.935
ecf_4R402	1.069	0.935
ecf_1R137	1.067	0.937
ecf_3R321	1.065	0.939
ecf_3R304	1.063	0.941
ecf_1R102	1.063	0.941
ecf_1R114	1.062	0.942
ecf_5R532	1.060	0.943
ecf_3R316	1.059	0.944
ecf_1R124	1.057	0.946
ecf_2R208	1.056	0.947
ecf_1R125	1.054	0.949
ecf_5R528	1.053	0.950

term	vif	tolerance
ecf_2R216	1.051	0.951
ecf_7R707	1.051	0.951
poor	1.050	0.952
ecf_7R724	1.049	0.954
ecf_3R324	1.048	0.954
ecf_3R326	1.046	0.956
ecf_5R533	1.046	0.956
ecf_3R309	1.045	0.957
ecf_3R311	1.044	0.958
ecf_5R509	1.043	0.959
ecf_6R601	1.041	0.960
good	1.041	0.961
ecf_3R308	1.040	0.962
ecf_6R612	1.039	0.963
ecf_5R534	1.037	0.965
ecf_1R133	1.032	0.969
ecf_5R507	1.032	0.969
ecf_6R608	1.031	0.970
ecf_2R218	1.029	0.971
ecf_5R531	1.029	0.972
ecf_1R131	1.026	0.975
ecf_6R606	1.024	0.976
ecf_1R107	1.023	0.978
ecf_4R416	1.022	0.979
ecf_7R715	1.020	0.980
ecf_7R722	1.020	0.980
ecf_5R523	1.019	0.981
ecf_7R720	1.019	0.982
ecf_6R614	1.019	0.982
ecf_3R301	1.019	0.982
ecf_3R318	1.018	0.983
ecf_4R418	1.017	0.983
ecf_5R510	1.017	0.983
ecf_2R226	1.017	0.984
ecf_5R505	1.016	0.984
very_poor_unsound	1.016	0.984
ecf_5R521	1.015	0.985
ecf_6R618	1.015	0.985
ecf_2R219	1.014	0.987
ecf_5R539	1.014	0.987
ecf_2R228	1.014	0.987
ecf_2R229	1.013	0.987
ecf_2R217	1.013	0.987
ecf_5R517	1.012	0.988
ecf_3R305	1.012	0.988
ecf_5R538	1.008	0.992
ecf_5R508	1.008	0.992
ecf_2R211	1.008	0.992
ecf_3R317	1.008	0.992
ecf_5R530	1.008	0.992
ecf_5R519	1.005	0.995
ecf_2R230	1.004	0.996
ecf_7R714	1.002	0.998
ecf_6R609	1.002	0.998

Table 16: VIF / GVIF

term	vif	tolerance
lnsbldsfint	16.639	0.060
largest	5.582	0.179
smallest	5.410	0.185
large	3.160	0.316
two_plus_baths	3.082	0.324
small	2.961	0.338
powder_room	2.561	0.391
yb_pre_1929	2.333	0.429
qabove_avg	2.128	0.470
ecf_7R705	2.116	0.472
qbelow_avg	2.113	0.473
ecf_2R214	2.066	0.484
fireplace	1.773	0.564
ln_landratio	1.727	0.579
yb_1946_to_1960	1.605	0.623
single_family1story	1.578	0.634
hot_water	1.543	0.648
ecf_7R704	1.474	0.679
ecf_4R410	1.470	0.680
ecf_2R201	1.455	0.687
ecf_7R719	1.453	0.688
ecf_3R319	1.453	0.688
ecf_4R406	1.419	0.705
ecf_2R215	1.393	0.718
ecf_2R212	1.388	0.721
ecf_1R104	1.384	0.723
ecf_1R109	1.373	0.728
ecf_4R405	1.359	0.736
ecf_2R203	1.348	0.742
ecf_2R205	1.336	0.749
ecf_2R221	1.330	0.752
ecf_2R222	1.315	0.760
ecf_3R303	1.311	0.763
ecf_3R307	1.307	0.765
ecf_2R213	1.304	0.767
ecf_3R323	1.300	0.769
ecf_1R117	1.291	0.775
ecf_4R408	1.290	0.775
ecf_1R105	1.288	0.776
ecf_4R417	1.286	0.777
ecf_6R620	1.280	0.781
ecf_7R711	1.278	0.782
ecf_4R413	1.278	0.782
ecf_5R504	1.268	0.789
ecf_3R320	1.253	0.798
ecf_4R411	1.250	0.800
yb_post_1961	1.245	0.803
ecf_6R611	1.239	0.807
ecf_2R220	1.238	0.808
ecf_7R708	1.236	0.809
ecf_1R108	1.236	0.809
ecf_2R223	1.233	0.811
crawl	1.232	0.812
ecf_2R225	1.231	0.813
ecf_2R224	1.227	0.815

term	vif	tolerance
ecf_7R709	1.221	0.819
ecf_1R121	1.216	0.822
ecf_1R128	1.212	0.825
ecf_1R116	1.202	0.832
ecf_7R710	1.200	0.833
ecf_3R310	1.200	0.834
ecf_7R718	1.197	0.835
ecf_4R401	1.190	0.840
ecf_1R118	1.184	0.844
ecf_1R136	1.183	0.845
ecf_4R404	1.182	0.846
ecf_3R312	1.181	0.846
ecf_6R604	1.178	0.849
ecf_5R502	1.178	0.849
ecf_4R403	1.173	0.852
ecf_1R122	1.170	0.855
ecf_1R135	1.158	0.864
wall	1.152	0.868
garage	1.151	0.869
ecf_1R126	1.148	0.871
ecf_5R501	1.145	0.873
ecf_3R322	1.143	0.875
ecf_1R110	1.143	0.875
ecf_3R315	1.142	0.876
ecf_7R717	1.142	0.876
ecf_3R325	1.142	0.876
ecf_3R313	1.141	0.876
ecf_7R701	1.140	0.877
ecf_4R422	1.138	0.879
ecf_7R713	1.136	0.880
ecf_1R119	1.136	0.880
ecf_3R306	1.133	0.883
ecf_1R129	1.126	0.888
half_duplex2story	1.124	0.890
ecf_7R723	1.122	0.891
ecf_4R421	1.122	0.892
central_air	1.113	0.898
ecf_5R503	1.113	0.899
ecf_1R112	1.112	0.899
ecf_4R409	1.109	0.902
ecf_1R123	1.102	0.908
slab	1.101	0.908
ecf_4R415	1.101	0.908
ecf_2R202	1.100	0.909
ecf_1R106	1.097	0.912
ecf_6R602	1.096	0.912
ecf_7R712	1.095	0.913
ecf_2R209	1.094	0.914
ecf_5R529	1.093	0.915
ecf_4R419	1.091	0.917
ecf_7R716	1.088	0.919
fair	1.088	0.919
ecf_7R703	1.085	0.921
ecf_2R210	1.084	0.922
ecf_5R514	1.082	0.924
ecf_6R605	1.082	0.924
ecf_4R407	1.079	0.927

term	vif	tolerance
ecf_4R412	1.078	0.928
ecf_3R314	1.077	0.929
ecf_1R127	1.074	0.931
ecf_6R603	1.074	0.931
ecf_1R120	1.072	0.933
ecf_1R113	1.071	0.934
ecf_5R506	1.070	0.935
ecf_7R706	1.069	0.935
ecf_4R402	1.069	0.935
ecf_1R137	1.067	0.937
ecf_3R321	1.065	0.939
ecf_5R532	1.064	0.940
ecf_3R304	1.063	0.940
ecf_1R102	1.063	0.941
ecf_1R114	1.062	0.941
ecf_3R316	1.059	0.944
ecf_2R208	1.057	0.946
ecf_1R124	1.057	0.946
ecf_2R216	1.054	0.949
ecf_5R528	1.053	0.949
ecf_1R125	1.053	0.950
ecf_7R707	1.052	0.951
poor	1.051	0.951
ecf_7R724	1.049	0.953
ecf_3R324	1.049	0.954
ecf_5R533	1.048	0.955
ecf_3R326	1.047	0.955
ecf_5R509	1.046	0.956
ecf_3R309	1.045	0.956
ecf_3R311	1.044	0.958
ecf_6R601	1.042	0.960
good	1.041	0.961
ecf_6R612	1.039	0.962
ecf_3R308	1.039	0.962
ecf_5R534	1.037	0.964
ecf_5R507	1.034	0.968
ecf_1R133	1.033	0.968
ecf_6R608	1.032	0.969
ecf_2R218	1.030	0.971
ecf_5R531	1.030	0.971
ecf_1R131	1.026	0.975
ecf_6R606	1.025	0.976
ecf_1R107	1.023	0.978
ecf_4R416	1.022	0.979
ecf_7R715	1.021	0.980
ecf_7R722	1.020	0.980
ecf_7R720	1.019	0.981
ecf_5R523	1.019	0.981
ecf_6R614	1.019	0.981
ecf_3R301	1.019	0.982
ecf_3R318	1.018	0.982
ecf_5R505	1.018	0.983
ecf_4R418	1.017	0.983
ecf_5R510	1.017	0.983
very_poor_unsound	1.016	0.984
ecf_5R521	1.016	0.985
ecf_6R618	1.015	0.985

term	vif	tolerance
ecf_2R219	1.014	0.986
ecf_5R539	1.014	0.986
ecf_2R229	1.014	0.986
ecf_2R226	1.014	0.986
ecf_2R228	1.014	0.986
ecf_5R535	1.014	0.987
ecf_3R305	1.013	0.988
ecf_5R517	1.012	0.988
ecf_5R508	1.009	0.991
ecf_5R538	1.009	0.992
ecf_2R211	1.008	0.992
ecf_3R317	1.008	0.992
ecf_5R530	1.008	0.992
ecf_5R519	1.005	0.995
ecf_2R230	1.004	0.996
ecf_7R714	1.003	0.997
ecf_6R609	1.003	0.997

